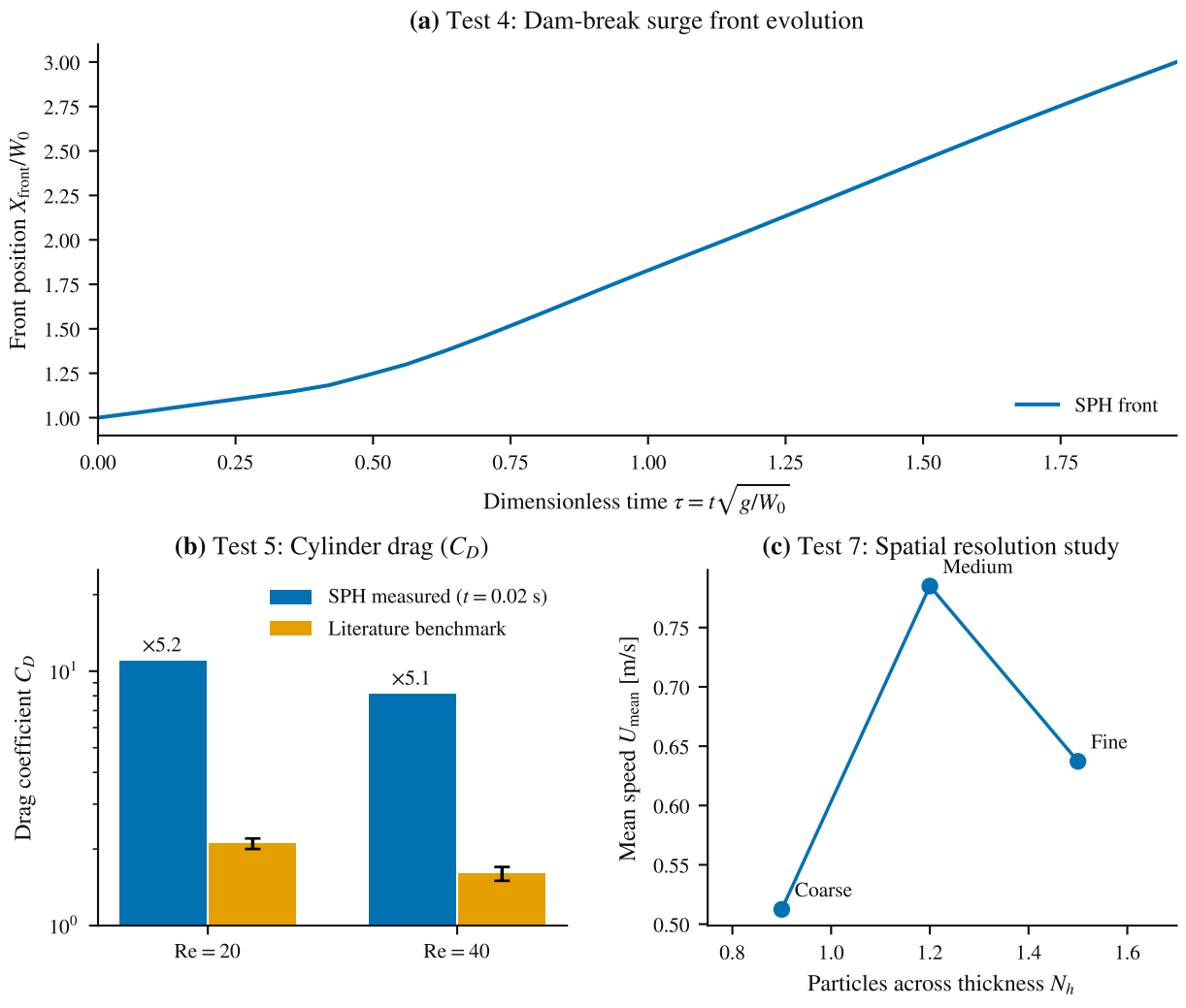
(a) Test 4: Dam-break surge front evolution
Front position X_{front}/W_0
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
SPH front
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Dimensionless time $\tau = t\sqrt{g/W_0}$
(b) Test 5: Cylinder drag (C_D)
SPH measured ($t = 0.02$ s)
Literature benchmark
Drag coefficient C_D
10^1
10^0
×5.2
×5.1
Re = 20
Re = 40
(c) Test 7: Spatial resolution study
Mean speed U_{mean} [m/s]
0.75
0.70
0.65
0.60
0.55
0.50
Medium
Fine
Coarse
0.8
1.0
1.2
1.4
1.6
Particles across thickness N_h